

WMC Question Writing Work Sample

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Part I

Sample Proof Round Questions

1 The Cauchy Functional Equation

Problem 1a. Prove that for $f: \mathbb{Q} \rightarrow \mathbb{Q}$,

$$f(x) = cx \iff f(x+y) = f(x) + f(y)$$

for any $x, y \in \mathbb{Q}$. (This was a question in the WMC Finals already, but I include it here for the sake of completeness.)

Solution 1a. We begin by noting that if $f(x) = cx$ for all $x \in \mathbb{R}$, then, trivially,

$$\begin{aligned} f(x+y) &= c(x+y) \\ &= cx + cy \\ &= f(x) + f(y) \end{aligned}$$

for all $x, y \in \mathbb{R}$. That is one direction of the implication done. Now, we just need to handle the other one.

We proceed by proving that if

$$f(x+y) = f(x) + f(y)$$

for any $x, y \in \mathbb{Q}$, then $f(x) = cx$ for any $x \in \mathbb{Q}$. We do this by first doing induction over the natural numbers and then extending our result to all integers.

Let $f(1) = c$. Also, let $P(x)$ refer to the proposition that $f(x) = cx$. Then, we notice that assuming $P(x)$,

$$\begin{aligned} f(x+1) &= f(x) + f(1) \\ &= cx + c \\ &= c(x+1), \end{aligned}$$

i.e.

$$P(x) \implies P(x+1).$$

As our base case is $P(1)$, by induction, we have $P(n)$ for all $n \in \mathbb{Z}^+$.

We proceed to handling positive rational numbers. We start with reciprocals of positive integers. Notice that for $n \in \mathbb{Z}^+$,

$$\begin{aligned} c &= f(1) \\ &= f\left(n \cdot \frac{1}{n}\right) \\ &= f\left(\frac{1}{n} + \cdots + \frac{1}{n}\right) \text{ (} n \text{ copies)} \\ &= f\left(\frac{1}{n}\right) + \cdots + f\left(\frac{1}{n}\right) \text{ (} n \text{ copies)} \\ &= n \cdot f\left(\frac{1}{n}\right). \end{aligned}$$

Hence,

$$f\left(\frac{1}{n}\right) = c \cdot \frac{1}{n}.$$

Moving on to covering all rationals, notice that for $p, q \in \mathbb{Z}^+$,

$$\begin{aligned} f\left(\frac{p}{q}\right) &= f\left(p \cdot \frac{1}{q}\right) \\ &= f\left(\frac{1}{q} + \cdots + \frac{1}{q}\right) \text{ (} p \text{ copies)} \\ &= f\left(\frac{1}{q}\right) + \cdots + f\left(\frac{1}{q}\right) \text{ (} p \text{ copies)} \\ &= p \cdot f\left(\frac{1}{q}\right) \\ &= c \cdot \frac{p}{q}. \end{aligned}$$

Hence,

$$f(x) = cx$$

for all $x \in \mathbb{Q}^+$.

Proceeding to extend our result to rational numbers of all signs, notice that

$$f(0) + f(0) = f(0+0) = f(0).$$

Hence, $f(0) = 0$. Then, notice that as $f(x) = cx$ for all $x \in \mathbb{Q}^+$.

$$\begin{aligned} f(-x+x) &= f(-x) + f(x) \\ f(0) &= f(-x) + cx \\ 0 &= f(-x) + cx \\ f(-x) &= c(-x). \end{aligned}$$

Hence,

$$f(x) = cx$$

for all $x \in \mathbb{Q}$. We will finish the problem using this intermediate result and the idea of continuity.

Problem 1b. Hence or otherwise, prove that if $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous, then

$$f(x) = cx \iff f(x + y) = f(x) + f(y)$$

for any $x, y \in \mathbb{R}$. (Hint: generalize your result from earlier using some calculus.)

Solution 1b. *The key intuition is that all real numbers are limits of sequences of rational numbers.* Hence, we will try to construct a sequence that approaches x for every $x \in \mathbb{R}$. Let

$$x_n = \frac{\lfloor nx \rfloor}{n}.$$

Notice that

$$nx - 1 \leq \lfloor nx \rfloor \leq nx.$$

Hence,

$$\begin{aligned} \frac{nx - 1}{n} &\leq \frac{\lfloor nx \rfloor}{n} \leq \frac{nx}{n} \\ x - \frac{1}{n} &\leq x_n \leq x. \end{aligned}$$

From the Squeeze Theorem, as

$$\lim_{n \rightarrow \infty} \left(x - \frac{1}{n} \right) = \lim_{n \rightarrow \infty} x = x,$$

we have

$$x = \lim_{n \rightarrow \infty} x_n.$$

Therefore, as f is continuous,

$$\begin{aligned} f(x) &= f\left(\lim_{n \rightarrow \infty} x_n\right) \\ &= \lim_{n \rightarrow \infty} f(x_n) \text{ (this allows us to use } f(x) = cx \text{ for all } x \in \mathbb{Q}) \\ &= \lim_{n \rightarrow \infty} c \cdot x_n \\ &= c \cdot \lim_{n \rightarrow \infty} x_n \\ &= cx \end{aligned}$$

for all $x \in \mathbb{R}$.

2 Inequalities Through Calculus

Problem 2a. Prove that, for any f where $f'' \geq 0$ and where the following inputs are defined,

$$f(\bar{x}) \leq \sum_{i=1}^n a_i f(x_i),$$

where $\{a_i\}_{i=1}^n$ is a sequence of positive real numbers and $\{x_i\}_{i=1}^n$ is a sequence of real numbers,

$$\sum_{i=1}^n a_i = 1,$$

and

$$\bar{x} = \sum_{i=1}^n a_i x_i.$$

(This inequality is known as Jensen's Inequality.)

Solution 2a. We begin by considering an individual x_i and $\bar{x}$. Notice that either $x_i > \bar{x}$, $x_i < \bar{x}$, or $x_i = \bar{x}$. Also, notice that as $f'' \geq 0$, f' is increasing. (Not monotonically, though.) *The key to this problem is to use the fact that f' is increasing alongside the Mean Value Theorem. (It can also be thought of as a consideration of the area under f' , which is f .)*

In the case that $\bar{x} > x_i$, from the Mean Value Theorem, for some c such that $x_i < c < \bar{x}$,

$$\frac{f(\bar{x}) - f(x_i)}{\bar{x} - x_i} = f'(c).$$

Since f' is increasing,

$$\frac{f(\bar{x}) - f(x_i)}{\bar{x} - x_i} = f'(c) \leq f'(\bar{x}).$$

Since $\bar{x} - x_i > 0$,

$$f(\bar{x}) - f(x_i) \leq f'(\bar{x})(\bar{x} - x_i).$$

In the case that $\bar{x} < x_i$, from the Mean Value Theorem, for some c such that $\bar{x} < c < x_i$,

$$\frac{f(\bar{x}) - f(x_i)}{\bar{x} - x_i} = f'(c).$$

Since f' is increasing,

$$\frac{f(\bar{x}) - f(x_i)}{\bar{x} - x_i} = f'(c) \geq f'(\bar{x}).$$

Since $\bar{x} - x_i < 0$,

$$f(\bar{x}) - f(x_i) \leq f'(\bar{x})(\bar{x} - x_i).$$

In the case that $\bar{x} = x_i$,

$$f(\bar{x}) - f(x_i) \leq f'(\bar{x})(\bar{x} - x_i).$$

is trivially true because both the left-hand and right-hand sides of the inequality are equal to zero.

Now, multiplying both sides by a_i and summing over i from 1 to n inclusive,

$$\sum_{i=1}^n a_i (f(\bar{x}) - f(x_i)) \leq \sum_{i=1}^n a_i (f'(\bar{x})(\bar{x} - x_i))$$

$$f(\bar{x}) - \sum_{i=1}^n a_i f(x_i) \leq f'(\bar{x})(\bar{x} - \bar{x}) \leq 0$$

$$f(\bar{x}) \leq \sum_{i=1}^n a_i f(x_i).$$

Problem 2b. Let

$$\mathcal{P}(x) = \begin{cases} \left(\frac{a_1^x + a_2^x + \cdots + a_n^x}{n} \right)^{\frac{1}{x}} & \text{if } x \neq 0, \\ (a_1 a_2 \cdots a_n)^{\frac{1}{n}} & \text{if } x = 0, \end{cases}$$

where $a_1, \dots, a_n$ are positive real numbers. Prove that $\mathcal{P}(r) \geq \mathcal{P}(s)$ if $r > s$.

Solution 2b. Notice that we are to prove that $\mathcal{P}(x)$ is an increasing function, that is, that $\mathcal{P}'(x) \geq 0$. Also, since $\mathcal{P}$ has two definitions, we would like to prove that $\lim_{x \rightarrow 0} \mathcal{P}(x) = \mathcal{P}(0)$ so that we can show that $\mathcal{P}$ is continuous.

We begin by showing that $\mathcal{P}'(x) \geq 0$ when $x \neq 0$. *The key to this is to first consider $\ln \mathcal{P}(x)$, to get rid of the exponent $\frac{1}{x}$, and take its derivative. Then, we manipulate our inequality into a form that can be finished off with Jensen's Inequality; we do this by doing some algebraic cleanup and then using a substitution to get rid of the exponent in a_i^x .* When $x \neq 0$,

$$\ln \mathcal{P}(x) = \frac{1}{x} \ln \left(\frac{\sum_{i=1}^n a_i^x}{n} \right).$$

Taking the derivative of both sides,

$$\begin{aligned} \frac{\mathcal{P}'(x)}{|\mathcal{P}(x)|} &= -\frac{1}{x^2} \ln \left(\frac{\sum_{i=1}^n a_i^x}{n} \right) + \frac{1}{x} \cdot \frac{\sum_{i=1}^n \ln(a_i) \cdot a_i^x}{|\sum_{i=1}^n a_i^x|} \\ &= -\frac{1}{x^2} \ln \left(\frac{\sum_{i=1}^n a_i^x}{n} \right) + \frac{1}{x} \cdot \frac{\sum_{i=1}^n \ln(a_i) \cdot a_i^x}{\sum_{i=1}^n a_i^x}. \end{aligned}$$

Hence,

$$\begin{aligned}
& \mathcal{P}'(x) \geq 0 \\
& \iff -\frac{1}{x^2} \ln \left(\frac{\sum_{i=1}^n a_i^x}{n} \right) + \frac{1}{x} \cdot \frac{\sum_{i=1}^n \ln(a_i) \cdot a_i^x}{\sum_{i=1}^n a_i^x} \geq 0 \\
& \iff \frac{1}{x} \cdot \frac{\sum_{i=1}^n \ln(a_i) \cdot a_i^x}{\sum_{i=1}^n a_i^x} \geq \frac{1}{x^2} \ln \left(\frac{\sum_{i=1}^n a_i^x}{n} \right) \\
& \iff x \cdot \frac{\sum_{i=1}^n \ln(a_i) \cdot a_i^x}{\sum_{i=1}^n a_i^x} \geq \ln \left(\frac{\sum_{i=1}^n a_i^x}{n} \right) \\
& \iff x \sum_{i=1}^n \ln(a_i) \cdot a_i^x \geq \left(\sum_{i=1}^n a_i^x \right) \ln \left(\frac{\sum_{i=1}^n a_i^x}{n} \right).
\end{aligned}$$

Now, let $A_i = a_i^x$. Because a_i is positive, A_i is positive too. Hence, our inequality is equivalent to

$$\begin{aligned}
& x \sum_{i=1}^n \ln \left(A_i^{\frac{1}{x}} \right) \cdot A_i \geq \left(\sum_{i=1}^n A_i \right) \ln \left(\frac{\sum_{i=1}^n A_i}{n} \right) \\
& \iff \sum_{i=1}^n A_i \ln(A_i) \geq \left(\sum_{i=1}^n A_i \right) \ln \left(\frac{\sum_{i=1}^n A_i}{n} \right).
\end{aligned}$$

Let $f(x) = x \ln x$ and $\bar{a} = \frac{\sum_{i=1}^n A_i}{n}$. Then, this inequality is equivalent to

$$\begin{aligned}
& \sum_{i=1}^n f(A_i) \geq \sum_{i=1}^n \left(n \cdot f(\bar{a}) \right) \\
& \iff \frac{1}{n} \sum_{i=1}^n f(A_i) \geq \sum_{i=1}^n f(\bar{a}).
\end{aligned}$$

Since

$$f''(x) = \frac{d}{dx}(\ln x + 1) = \frac{1}{x} > 0,$$

f is concave up, so the inequality we were considering above is true by Jensen's Inequality. Hence,

$$\mathcal{P}'(x) \geq 0$$

for $x \neq 0$. In order to show that $\mathcal{P}(r) \geq \mathcal{P}(s)$ if $r > s$, we also have to show that $\mathcal{P}$ is continuous at $x = 0$. *The motivation for this is that we are really almost done, but need to show that our inequality is not ruined by a hole at $x = 0$.*

We now proceed to show that $\lim_{x \rightarrow 0} \mathcal{P}(x) = \mathcal{P}(0)$. As before, it is helpful to consider $\ln \mathcal{P}(x)$. For $x \neq 0$,

$$\ln \mathcal{P}(x) = \frac{1}{x} \ln \left(\frac{\sum_{i=1}^n a_i^x}{n} \right),$$

so

$$\lim_{x \rightarrow 0} \ln \mathcal{P}(x) = \lim_{x \rightarrow 0} \frac{1}{x} \ln \left(\frac{\sum_{i=1}^n a_i^x}{n} \right).$$

Notice that

$$\begin{aligned} \lim_{x \rightarrow 0} \ln \left(\frac{\sum_{i=1}^n a_i^x}{n} \right) &= \lim_{x \rightarrow 0} \ln \left(\frac{\sum_{i=1}^n a_i^0}{n} \right) \\ &= \lim_{x \rightarrow 0} \ln \left(\frac{n}{n} \right) \\ &= \ln 1 \\ &= 0, \end{aligned}$$

so we have a $\frac{0}{0}$ indeterminate form, allowing us to use l'Hopital's Rule. Hence,

$$\begin{aligned} \lim_{x \rightarrow 0} \ln \mathcal{P}(x) &= \lim_{x \rightarrow 0} \frac{\sum_{i=1}^n (\ln a_i) a_i^x}{\sum_{i=1}^n a_i^x} \\ &= \lim_{x \rightarrow 0} \frac{\sum_{i=1}^n \ln a_i}{n} \\ &= \lim_{x \rightarrow 0} \ln(a_1 a_2 \cdots a_n)^{\frac{1}{n}} \\ &= \ln(a_1 a_2 \cdots a_n)^{\frac{1}{n}} \\ &= \ln \mathcal{P}(0). \end{aligned}$$

Hence,

$$\lim_{x \rightarrow 0} \mathcal{P}(x) = \mathcal{P}(0).$$

3 Euler's Totient Formula Through Probability

Problem 3. Note that every $n \in \mathbb{Z}^+$ other than 1 has a unique prime factorization, which can be written as

$$n = \prod_{i=1}^M p_i^{k_i}$$

for some sequence of primes $\{p_i\}_{i=1}^M$ and some sequence of positive integers $\{k_i\}_{i=1}^M$. Let $\phi(n)$ be the number of integers I such that $1 \leq I \leq n$ and $\gcd(I, n) = 1$.

Solution 3. We solve this problem by considering the probability that some randomly chosen integer I such that $1 \leq I \leq n$ satisfies $\gcd(I, n) = 1$ in order to bypass the classical proof. We break $\gcd(I, n) = 1$ down into cases by considering the idea that the probabilities that any of the prime divisors p_i of n divide I are independent from each other.

Before that, we will prove a lemma: if $a \mid n$, then $P(a \mid I) = \frac{1}{a}$ for some randomly chosen integer I such that $1 \leq I \leq n$. Let $n = ab$ for some $b \in \mathbb{Z}^+$. Then,

$$\begin{aligned} P(a \mid I) &= \frac{\#(a \mid I)}{n} \\ &= \frac{\#(I = a, I = 2a, I = 3a, \dots, I = ab = n)}{n} \\ &= \frac{b}{n} \\ &= \frac{1}{a}. \end{aligned}$$

Therefore,

$$\begin{aligned} P(\gcd(I, n) = 1) &= P((p_1 \nmid I) \cap (p_2 \nmid I) \cap \dots \cap (p_M \nmid I)) \\ &= P(\text{lcm}(p_1, p_2, \dots, p_M) \nmid I) \\ &= P(p_1 \nmid I) \cdot P(p_2 \nmid I) \cdots P(p_M \nmid I) \\ &\quad (\text{independent by the Chinese Remainder Theorem}) \\ &= \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_M}\right). \end{aligned}$$

Hence, as

$$\frac{\phi(n)}{n} = P(\gcd(I, n) = 1) = \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_M}\right),$$

we have

$$\phi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_M}\right).$$

4 Sizes of Infinity: $|\mathbb{Z}^+| = |\mathbb{Q}^+|$

Remark. In this problem, we determine which infinite sets have the same cardinality, or size, as each other. To show that $|A| = |B|$ – that set A has the same size as set B – we can show that every $a \in A$ maps to a unique $b \in B$, and every $b \in B$ maps to a unique $a \in A$. We can specify the mapping by writing functions with the domain A and codomain B . Indeed, functions are *defined* as mappings between sets.

Problem 4a. Write a function s that establishes a bijection between $\mathbb{Z}^+$ (positive integers) and $\mathbb{Z}$ (integers), i.e. show that $|\mathbb{Z}^+| = |\mathbb{Z}|$.

Solution 4a. Let $s(n) = (-1)^n \lfloor \frac{n}{2} \rfloor$. Notice that s maps every $n \in \mathbb{Z}^+$ to a unique $m \in \mathbb{Z}$. This is because there are two n with the same $\lfloor \frac{n}{2} \rfloor$, but the

parity of n determines the sign of m . Similarly, solving the equation $s(n) = m$ where $m \in \mathbb{Z}$ yields a unique $n \in \mathbb{Z}$. This is because for $m \neq 0$, there are two n such that $\lfloor \frac{n}{2} \rfloor = |m|$, and the sign of m determines the parity of n . If $m = 0$, then $n = 1$. ($n = 1$ only happens for $m = 0$, too.) Hence, s is a bijection between $\mathbb{Z}^+$ and $\mathbb{Z}$.

Problem 4b. Hence or otherwise, show that there exists a bijection between the positive integers and the positive rationals, i.e. that $|\mathbb{Z}^+| = |\mathbb{Q}^+|$.

Solution 4b. *The key idea is to note that both positive integers and positive rational numbers have unique prime factorizations. We can use our function to establish a bijection between them by applying it to the exponents of the primes in the factorizations of the positive integers.* Notice that every $n \in \mathbb{Z}^+$ can be written as

$$n = p_1^{k_1} p_2^{k_2} \cdots$$

where $p_1, p_2, \dots$ are primes and $k_1, k_2, \dots \in \mathbb{Z}^+$. Therefore, let

$$f(n) = \begin{cases} 1 & \text{if } n = 1, \\ p_1^{s(k_1)} p_2^{s(k_2)} \cdots & \text{if } n \neq 1. \end{cases}$$

We see that f maps every positive rational number to a unique positive integer and every positive integer to a unique rational number because s is a bijection between positive integers and integers.